

# Phil Legg

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## Experience

### University of the West of England (UWE Bristol), Professor of Cyber Security

Bristol, UK

Jan 2022 – present  
4 years 4 months

- Full Professor in Cyber Security in recognition of 'Advancing Teaching and Learning' (primary domain) and for 'Research with Impact' (secondary domain).
- Successful research grant income with over £2.1M directed awarded from external funding sources, and over £3.7M in total (see projects page for further details).
- Published 90+ academic articles as of April 2026, in high-quality journals and conference venues (see publications page for further details).
- Supervision of PhD student research (4 completions, 6 current
- March 2026), as well as supporting post-doctoral research fellows (2 former, 2 current
- March 2026) (see projects page for further details).
- Experience of supervising 100+ student research projects across undergraduate and postgraduate level, as well as supporting extra curricula paid student research and consultancy work (see projects page for further details).
- Leading outreach initiatives and upskilling educators and industry professionals in cyber security (see teaching page for further details).

### University of the West of England (UWE Bristol), Co-Director of UWEcyber Academic Centre of Excellence in Cyber Security Education (NCSC Certified)

Bristol, UK

Oct 2020 – present  
5 years 8 months

- Co-Director of UWEcyber 'Gold' NCSC Academic Centre of Excellence in Cyber Security Education (ACE-CSE) that brings together our teaching and learning, external engagement, outreach, research and knowledge exchange in Cyber Security, with a multi-disciplinary perspective involving staff across the entire University, a regional network of educators with a passion to upskill cyber security education, and a regular guest speaker seminar series that has attracted presenters from across the globe.

### University of the West of England (UWE Bristol), Programme Leadership Team - BSc Cyber Security and Digital Forensics

Bristol, UK

July 2024 – present  
1 year 11 months

- I joined the BSc Cyber Security and Digital Forensics programme leadership team in 2024, to support and mentor a new leadership team, and to further promote opportunities of external industry engagement for our undergraduate students, including the continued organisation of guest speakers as well as new industry collaboration with teaching.

### University of the West of England (UWE Bristol), CSRC Cyber Security Research Theme Lead

Bristol, UK

Aug 2019 – present  
6 years 10 months

- The Computer Science Research Centre (CSRC) at UWE is made up of 5 themes: Artificial Intelligence, Cyber Security, Data Science, Expressive Computer Interaction, and Smart Cities. I serve as Cyber Security Research Theme Lead since the reorganisation of the CSRC in 2019, supporting academic staff and PhD researchers to develop cyber security research across infrastructure security, software security, AI/data security, and cyber crime domains.

### University of the West of England (UWE Bristol), Programme Leader - MSc Cyber Security (NCSC Certified)

Bristol, UK

Jan 2018 – July 2024  
6 years 8 months

- Former Programme Leader of the NCSC-Certified MSc Cyber Security, having grown the programme from its initial delivery for 13 students in 2018 to as many as 150 students across two intakes in an academic year, and achieving a PTES overall student satisfaction score of 92% (2024).

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| <b>University of the West of England (UWE Bristol)</b> , Associate Professor of Cyber Security                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Bristol, UK<br>Sept 2018 – Jan 2022<br>3 years 4 months |
| <b>University of the West of England (UWE Bristol)</b> , Senior Lecturer in Computer Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Bristol, UK<br>Feb 2015 – Sept 2018<br>3 years 8 months |
| <b>Cabinet Office</b> , Member of the UK Government Cyber Advisory Board (GCAB)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | London, UK<br>Mar 2025 – present<br>1 year 2 months     |
| <b>CSE Connect</b> , Co-Founder / Co-Director <ul style="list-style-type: none"> <li>A National Cyber Security Education network to bring together academia, industry and government. Supported by the UK National Cyber Security Centre, the network hosts regular events as well as an annual conference, that promotes collaboration and sharing to enhance cyber security education, both within the UK and working with global partners. (Formerly CISSEUK, a UK chapter of CISSE)</li> <li>the Colloquium of Information Systems Security Education)</li> </ul>                                                                                                                                                                                                                                                                                                                                           | UK<br>July 2019 – present<br>6 years 11 months          |
| <b>University of Oxford</b> , Post-Doctoral Research Associate <ul style="list-style-type: none"> <li>Cyber Security Centre, Department of Computer Science</li> <li>Research project: ‘Corporate Insider Threat Detection (CITD)’ funded by the Centre for the Protection of National Infrastructure (CPNI).</li> <li>Lectureship post at St. Catherine College, teaching Computer Graphics.</li> <li>Lectureship post at Worcester College, tutoring International students in Computer Graphics</li> <li>Teaching at the Department of Computer Science includes classes for the Visual Analytics postgraduate course, and both tutorials and practical labs for the Computer Graphics course.</li> <li>Supervision of postgraduate projects in the Centre for Doctoral Training in Cyber Security</li> </ul>                                                                                                | Oxford, UK<br>Mar 2013 – Feb 2015<br>1 year 11 months   |
| <b>Swansea University</b> , Post-Doctoral Research Assistant <ul style="list-style-type: none"> <li>Department of Computer Science</li> <li>Research project: ‘Sports Video Visualization’ funded by the Welsh Assembly Government.</li> <li>Guest lecturer on iOS Software Development and Sports Video Visualisation</li> <li>‘Inspirational Speaker’ and iOS Schools Workshop Lead for Technocamps Wales.</li> <li>Supervision of undergraduate and postgraduate projects.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                        | Swansea, UK<br>July 2010 – Mar 2013<br>2 years 9 months |
| <b>Freelance</b> , Software Developer <ul style="list-style-type: none"> <li>Desktop and Mobile (iOS, Android, Blackberry) development.</li> <li>Over 40,000 paid downloads on the iOS App Store worldwide since July 2011, with Tide Times being recognised in the Stuff magazine ‘200 best apps of all time’, and also achieving #1 in the ‘Top Paid Weather’ category (Summer 2011).</li> <li>PocketCoder used by top sports clubs worldwide including Manchester United FC, Welsh Rugby Union, Ireland RFU, Barnsley FC, Bolton FC, QPR, and the Football Association.</li> <li>Desktop software includes collaborative development with University Hospital Wales and NHS, producing a 3D stereoscopic imaging system for ophthalmic clinical assessment based on PhD research.</li> <li>Proficient in variety of development languages: Python, Javascript, Java, C++, C#, Objective-C, Swift.</li> </ul> | UK<br>May 2010 – Feb 2015<br>4 years 10 months          |
| <b>Cardiff University</b> , Software Developer <ul style="list-style-type: none"> <li>Development of research-led Android mobile application, GeoCaption.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | UK<br>Apr 2010 – July 2010<br>4 months                  |

## Education

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|            | <b>University of the West of England (UWE Bristol)</b> , Postgraduate Certificate in Teaching and Learning for Higher Education                                                                                | Bristol, UK<br>2015 – 2016 |
| <b>PhD</b> | <b>Cardiff University, UK</b> , Computer Science <ul style="list-style-type: none"><li>Multi-modal Retinal Imaging: Improving Accuracy and Efficiency of Image Registration using Mutual Information</li></ul> | Cardiff, UK<br>2006 – 2010 |
| <b>BSc</b> | <b>Cardiff University, UK</b> , Computer Science <ul style="list-style-type: none"><li>Online Multi-player Chinese Checkers Game with Intelligent Computer Opponent</li></ul>                                  | Cardiff, UK<br>2003 – 2006 |

## Publications

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|  | <b>Adversarially Resilient Federated Learning for Heterogeneous Edge Nodes in 5G Networks with Non-IID Data</b><br>Saniya Zafar, Phil Legg, Jonathan White, Ahmad Salman<br><a href="https://doi.org/10.1016/j.iot.2026.101919">10.1016/j.iot.2026.101919</a> (Internet of Things)                                                                      | Mar 2026 |
|  | <b>LOTL-hunter: Detecting multi-stage living-off-the-land attacks in cyber-physical systems using decision fusion techniques with digital twins</b><br>Carol Lo, Thu Yein Win, Zeinab Rezaeifar, Zaheer Khan, Phil Legg<br><a href="https://doi.org/10.1016/j.future.2026.108382">10.1016/j.future.2026.108382</a> (Future Generation Computer Systems) | Mar 2026 |
|  | <b>Advances in Teaching and Learning for Cyber Security Education</b><br>Phil Legg, Natalie Coull, Charles Clarke<br><a href="https://doi.org/10.1007/978-3-032-15105-6">10.1007/978-3-032-15105-6</a> (Springer)                                                                                                                                       | Jan 2026 |
|  | <b>Machine Learning and Data Analytics for Cyber Security</b><br>Phil Legg, Giorgio Giacinto<br><a href="https://doi.org/10.3390/books978-3-7258-3360-3">10.3390/books978-3-7258-3360-3</a> (MDPI)                                                                                                                                                      | Apr 2025 |

## Certificates

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| <b>CompTIA SecurityX CAS-005</b>                                        | July 2024 |
| <b>Chartered Institute of Information Security (CIISec) Full Member</b> | Oct 2022  |
| <b>CompTIA PenTest+ PT0-002</b>                                         | May 2021  |
| <b>Security, Compliance, and Identity Fundamentals SC-900</b>           | Apr 2021  |
| <b>Azure Data Fundamentals DP-900</b>                                   | Mar 2021  |
| <b>Azure AI Fundamentals AI-900</b>                                     | Feb 2021  |
| <b>Azure Fundamentals AZ-900</b>                                        | Oct 2020  |
| <b>CompTIA Security+ SY0-501</b>                                        | July 2020 |

## Projects

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| <b>UWEcyber Research Laboratory strategic investment</b><br>Wallscourt Foundation (£1.5m).                                                               | Oct 2023 – Oct 2027 |
| <b>Cyber Sandpit - Linking cyber effect to mission objectives</b><br>DSTL Serapis Framework in collaboration with Trimetis Ltd. (Total £120k, UWE £50k). | Jan 2023 – Oct 2023 |

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| <b>Every human a sensor - Crowdsourcing Cyber Data</b><br>DSTL Serapis Framework in collaboration with Trimetis Ltd. (Total £80k, UWE £41k).                                                                                                                   | Jan 2023 – Oct 2023  |
| <b>Teaching CyBOK through Cyber Physical Systems</b><br>CyBOK (Cyber Security Body of Knowledge) and NCSC (Total £5k).                                                                                                                                         | Jan 2023 – June 2023 |
| <b>Autonomous Resilience in Cyber Defence</b><br>DSTL Serapis Framework in collaboration with Trimetis Ltd. (Total £30k, UWE £5k).                                                                                                                             | Sept 2022 – Nov 2022 |
| <b>Cyber Security and Data Governance for SMEs</b><br>National Cyber Security Centre (Total £40k).                                                                                                                                                             | Sept 2022 – Mar 2023 |
| <b>CyberWEST - Cyber Security Education for Schools</b><br>National Cyber Security Centre (Total £41k).                                                                                                                                                        | Sept 2022 – Mar 2023 |
| <b>Transform Suspicious Activity Reports (SARs) - UK First Technology Line of Defence</b><br>InnovateUK Smart Grant in collaboration with Synalogik Ltd. (Total £458k, UWE £37k).                                                                              | Sept 2022 – Feb 2024 |
| <b>Building a place-based approach to preventing online sexual abuse</b><br>Vice Chancellor's Challenge Fund 2022-23 (Total £25k internal).                                                                                                                    | May 2022 – Apr 2023  |
| <b>Turing Network Development Awards</b><br>Alan Turing Institute (Total £25k).                                                                                                                                                                                | Feb 2022 – Jan 2023  |
| <b>Cyber Security and Cyber Crime Cluster (CSC3)</b><br>UWE Expanding Research Excellence Scheme (Total £100k internal).                                                                                                                                       | Feb 2022 – July 2023 |
| <b>Cyber security analytics in telecommunications systems: Managing security and service in complex real-time 5G networks</b><br>PhD 50-50 studentship in collaboration with Ribbon Communications Ltd. (Total £35k external, £35k internal).                  | Jan 2022 – Dec 2024  |
| <b>Co-creation of innovative school projects for teachers to improve cyber security education</b><br>National Cyber Security Centre (Total £50k).                                                                                                              | Sept 2021 – Feb 2022 |
| <b>Teaching CyBOK Cyber-Physical Systems Security through Interactive Simulation</b><br>CyBOK / National Cyber Security Centre (Total £5k).                                                                                                                    | Jan 2021 – Apr 2021  |
| <b>Identifying and Tackling Counter-Terrorism Finances through Investigation Automation (SCOUT: A fully automated enhanced risk assessment engine)</b><br>InnovateUK Smart Grants (Total £490k).                                                               | Apr 2020 – Mar 2022  |
| <b>Cyber Security Assessment for Project CAVForth (An Autonomous Bus Service From Park &amp; Ride Across Forth Bridge to Edinburgh Park Train &amp; Tram Interchange)</b><br>Innovate UK and Centre for Connected and Autonomous Vehicles (CCAV) (Total £34k). | Jan 2020 – Dec 2020  |
| <b>Analysis and Visualisation of Ransomware and Malicious Network Propagation</b><br>National Cyber Security Centre Internship Award (Total £2.4k).                                                                                                            | May 2019 – Sept 2019 |
| <b>Penetration testing Consultancy</b><br>VQ Communications Ltd. (Total £9k).                                                                                                                                                                                  | Nov 2019 – July 2020 |

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| <b>Interactive Inspection of Adversarial Machine Learning for Collaborative Human-Machine Trust and Security in Data-Driven Military and Defence</b><br>PhD 50-50 studentship in collaboration with Techmodal Ltd. (Total £35k external, £35k internal). | Oct 2019 – Sept 2022  |
| <b>Shared Collaborative Project Planning in Secure Federated Learning Environments</b><br>PhD 50-50 studentship in collaboration with Techmodal Ltd. (Total £35k external, £35k internal).                                                               | Oct 2019 – Sept 2022  |
| <b>Travel Grant: Colloquium for Information Systems Security in Education 2019, Las Vegas, USA</b><br>National Cyber Security Centre (Total £2k).                                                                                                        | July 2019 – July 2019 |
| <b>Cyber Security in the Home: Detecting IoT Threats</b><br>National Cyber Security Centre Internship Award (Total £2.4k).                                                                                                                               | May 2019 – Sept 2019  |
| <b>Travel Grant: Colloquium for Information Systems Security in Education 2018, New Orleans, USA</b><br>National Cyber Security Centre (Total £2k).                                                                                                      | July 2018 – July 2018 |
| <b>HASTE: Human-centric Active-learning for decision Support in Threat Exploration</b><br>DSTL Defence and Security Accelerator (Total £85k).                                                                                                            | Feb 2018 – July 2018  |
| <b>Creating Machine Intelligence with Intelligent Interactive Visualisation</b><br>PhD 50-50 studentship in collaboration with Montvieux Ltd. (Total £35k external, £35k internal).                                                                      | Oct 2016 – Sept 2019  |
| <b>RicherPicture: Automated network defence through business and threat-led machine learning – Phase 2</b><br>DSTL Centre for Defence Enterprise (Total £280k).                                                                                          | Apr 2016 – Mar 2017   |
| <b>Enhanced personal situational awareness through network and systems visualization</b><br>UWE Bristol Vice-Chancellor's Early Career Researcher Development Awards (Total £13k internal).                                                              | Aug 2015 – July 2016  |
| <b>RicherPicture: Automated network defence through business and threat-led machine learning</b><br>DSTL Centre for Defence Enterprise (Total £108k).                                                                                                    | Apr 2015 – Sept 2016  |

## Awards

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| <b>Best Paper Award</b><br>IEEE International Conference on Cyber Situational Awareness, Data Analytics and Assessment (Cyber SA 2016)<br><a href="http://c-mric.org">c-mric.org</a> | July 2016 |
| <b>CSCT Researcher of the Year Award</b><br>Faculty of Environment and Technology, UWE Bristol<br><a href="http://www.uwe.ac.uk">www.uwe.ac.uk</a>                                   | July 2015 |
| <b>Best Paper Award</b><br>Managing Insider Security Threats (MIST) conference<br><a href="http://isyou.info/conf/mist14">isyou.info/conf/mist14</a>                                 | Nov 2014  |

## Invited Talks

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Transparency as The Foundation, Standards, Assurance, and Trust in Cyber Security (CYBERUK, 2026)

Trowbridge musician's warning after voice and face used in AI deep-fake romance scam (ITV News, 2026)